

# Data Science

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## Lecture 10: Projections and Dimensionality Reduction

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# Module 1

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## What a Projection Preserves

**Guiding Question:** *Reducing dimensions must lose information. Which information do you choose to keep?*

# Seeing High-Dimensional Data

A **geometric projection** maps high-dimensional data onto a low-dimensional space — usually 2-D or 3-D, so that it can be plotted and reasoned about.

## Linear projections

$$\mathbf{z} = \mathbf{P}\mathbf{x}$$

A matrix  $\mathbf{P}$  maps every point. Distances and angles are distorted in a predictable, global way the map is interpretable and invertible in structure.

## Non-linear projections

Preserve some **local** property — neighbourhoods, or pairwise distances — without a single global matrix. More faithful visually, harder to interpret.

**The central question of the chapter:** how do we find the *best* projection — and by which criterion is one projection better than another?

# What Is Necessarily Lost

Reducing dimensions **must** lose information — the question is only which.

## Variance

PCA keeps as much of it as a linear map can.

## Distances

MDS keeps pairwise distances as faithfully as possible.

## Guarantees

Random projection keeps distances within a provable error, cheaply.

A projection is defined by **what it chooses to preserve**. There is no best-in-general projection, only a best one for a stated criterion.

# The Surprising Result

**Johnson–Lindenstrauss lemma.** For any  $0 < \epsilon < 1$  and  $n$  points in any dimension, a projection onto

$$d = O\left(\frac{\log n}{\epsilon^2}\right)$$

dimensions exists that preserves all pairwise distances to within a factor  $1 \pm \epsilon$ .

The target dimension depends on the **number of points** and on the accuracy wanted — **not on the original dimensionality**. A million-dimensional dataset and a thousand-dimensional one need the same target dimension.

# Construction

1. Draw a  $d \times p$  matrix  $\mathbf{R}$  with entries sampled independently (Gaussian, or  $\pm 1$  with equal probability).
2. Project:  $\mathbf{z} = \frac{1}{\sqrt{d}} \mathbf{R} \mathbf{x}$ .

## Properties

No training, no data pass — the matrix does not depend on the data at all.

Extremely fast, trivially applied to new points, and it comes with a bound.

## Limits

The axes are meaningless, so the result is not interpretable. It preserves distances, not structure, and works best when  $d$  is not too small.

Random projection preserves distances **with a guarantee and without looking at the**

# Module 2

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## Principal Component Analysis

***Guiding Question:** PCA keeps the directions of largest variance. When is variance the wrong thing to keep?*

# Which Direction Matters?

Consider a cloud of points that is stretched along one diagonal. Projecting onto the **short** axis collapses everything into a blob projecting onto the **long** axis keeps the points apart.

**PCA's criterion:** keep the directions along which the data **varies most**. Variance is treated as information directions with little variance are treated as noise.

## 1st component

The direction of maximum variance.

## 2nd component

The direction of maximum remaining variance, **orthogonal** to the first.

## and so on

Each component orthogonal to all previous ones, each capturing less variance.

# What the Components Are

The components are ordered, so keeping the first  $m$  gives the **best possible  $m$ -dimensional linear approximation** in the least-squares sense.

Equivalently: PCA finds the linear subspace minimising the total squared distance from the points to their projections. Maximising retained variance and minimising reconstruction error are the **same** criterion.

PCA orders directions by **variance** and keeps the top ones. Its assumption — that variance is information — is what makes it powerful and what makes it fail.

# Covariance Is the Object to Diagonalise

For centred data  $\mathbf{X}$ , the **covariance matrix**

$$\mathbf{C} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{X}$$

holds the variance of each attribute on its diagonal and the covariance of each pair off it.

## Large diagonal entry

That attribute varies a lot — potentially informative.

## Large off-diagonal entry

Two attributes are **redundant**: one partly predicts the other.

The goal: find new axes in which the covariance matrix is **diagonal** — no redundancy left between components — and order them by variance.

# The Solution

The directions that diagonalise **C** are its **eigenvectors** the variance captured by each is the corresponding **eigenvalue**.

Sorting eigenvectors by descending eigenvalue gives the principal components, and the projection matrix is the top  $m$  of them.

## Orthogonal

Eigenvectors of a symmetric matrix are mutually orthogonal — the components are uncorrelated by construction.

## Ordered

$\lambda_1 \geq \lambda_2 \geq \dots$  ranks the directions by variance.

## Explained variance

$\lambda_i / \sum_j \lambda_j$  is the share of total variance in component  $i$ .

# The Procedure

1. **Centre** the data (and usually **standardise** it).
2. Compute the covariance matrix  $\mathbf{C}$ .
3. Compute its eigenvectors and eigenvalues.
4. Sort by descending eigenvalue keep the top  $m$ .
5. Project:  $\mathbf{Z} = \mathbf{X}\mathbf{W}_m$ .

Step 1 is not optional. Without standardisation, an attribute measured in metres dominates one measured in kilometres purely through its **unit** — PCA would report the choice of unit as the principal direction.

# Choosing the Number of Components

A **scree plot** shows eigenvalues in descending order; the usual reading is the same elbow heuristic as for choosing  $k$  in clustering, or a cumulative explained-variance threshold (80%, 95%).

## Benefits

Removes correlated and noisy directions  
speeds up downstream methods makes 2-D visualisation possible components are uncorrelated by construction.

## Drawbacks

Components are **linear combinations of all attributes**, so interpretability is lost. Only **linear** structure is captured. And variance is assumed to equal importance — which for classification can be false.

# Mini-Exercise

You run PCA on the iris dataset and the first two components explain 96% of the variance.

1. What does that number mean?
2. Does it mean the two components are good features for classifying species?
3. When could a **low-variance** direction be the important one?

# Mini-Exercise — Solution

PCA is centre  $\rightarrow$  covariance  $\rightarrow$  eigendecomposition  $\rightarrow$  keep the top components. It optimises **reconstruction**, not discrimination, and it must be standardised first.

# Mini-Exercise — Solution

1. A 2-D linear approximation reconstructs the data with only 4% of the total squared variation lost.

2. Not necessarily. PCA is **unsupervised** — it never saw the labels, and optimises reconstruction, not separability.

3. Whenever the classes differ along a direction with small spread — a thin but decisive offset. Discriminant analysis targets exactly that instead.

PCA is centre → covariance → eigendecomposition → keep the top components. It optimises **reconstruction**, not discrimination, and it must be standardised first.

# Points and Variables in One Picture

A **biplot** overlays two things on the same axes:

## The objects

Each data point, plotted by its scores on the first two components — an ordinary scatterplot of the projection.

## The variables

Each original attribute drawn as an **arrow**: its loading on the two components. Direction shows what the attribute contributes to, length shows how strongly.

## Arrows together

Those attributes are positively correlated.

## Arrows opposed

Negatively correlated.

## Arrows at 90°

Roughly uncorrelated.

A biplot is how you recover interpretability after PCA has destroyed it: the arrows say what



the anonymous components *mean* in terms of the original attributes.

The biplot answers "what do these axes mean?" by drawing the original attributes into the projected space — points and variables read together.

# Module 3

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## Beyond Variance

**Guiding Question:** *Some objects have no coordinates — only similarities. Can they still be mapped?*

# The Problem With Matching Words

A document is represented as a vector of term counts — the **term–document matrix**. Similarity is then the overlap of vocabulary, which fails twice:

## Synonymy

"car" and "automobile" never match, though the documents are about the same thing.

## Polysemy

"bank" matches across finance and rivers, though the documents are unrelated.

The vocabulary is the wrong space. We want a space of **concepts**, in which related terms lie close together.

# The Approach

Apply a **singular value decomposition** to the term–document matrix  $\mathbf{A}$ :

$$\mathbf{A} \approx \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top$$

and keep the  $k$  largest singular values. Documents and terms are both represented in the resulting  $k$ -dimensional **latent space**.

## Dimension reduction

Thousands of terms collapse to a few hundred latent dimensions.

## Synonyms merge

Terms that co-occur with the same contexts end up close together.

## Noise drops

Rare, idiosyncratic usage lives in the discarded small singular values.

SVD is PCA's underlying machinery applied to a rectangular matrix — the same "keep the strongest directions" idea, on term–document data.

LSA projects documents and terms into a shared **latent space** via SVD, so that similarity survives synonymy. It is dimensionality reduction used as a representation, not just as compression.

# A Different Criterion

PCA needs coordinates and preserves **variance**. MDS needs only a **distance matrix** and preserves **the distances themselves**.

Given pairwise distances  $d_{ij}$ , find low-dimensional points  $\mathbf{z}_1, \dots, \mathbf{z}_n$  minimising the **stress** 
$$S = \sum_{i < j} (\|\mathbf{z}_i - \mathbf{z}_j\| - d_{ij})^2$$

This is the projection method for objects that have **no coordinates at all** — only a similarity: documents, molecules, survey respondents, perceived similarity ratings.

# Classical MDS and Its Relatives

## Classical (metric) MDS

Solved in closed form by an eigendecomposition of the (double-centred) squared distance matrix. With Euclidean distances it is **equivalent to PCA**.

## Non-metric MDS

Preserves only the **rank order** of distances. Suited to ordinal similarity data — "A is more like B than like C" — where the numbers are not meaningful.

# Reading the Result

## Stress tells you the cost

A 2-D map of genuinely high-dimensional structure has high stress: the picture is a compromise, and the number says how big a one.

## Axes mean nothing

Unlike PCA components, MDS axes have no interpretation — only **relative positions** carry information. Rotations and reflections are equivalent.

MDS preserves **distances** rather than variance, works from a similarity matrix alone, and produces a map whose axes are meaningless but whose **neighbourhoods are faithful** — as far as the stress allows.